

Swasti Choubey

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Summary

Transitioning from production AI into empirical alignment research. I hold a Master's in Computer Science from UC Irvine, and my hands-on work in RAG systems, embedding evaluation, and AI security has sharpened a longstanding concern with how AI systems fail in ways that are hard to detect and easy to propagate. Research interests: evaluation reliability, information provenance, and interpretability - three angles on the same underlying problem of verifying rather than assuming a system does what we think it does.

Research Interests

Domains: *Evaluation Reliability, Information Provenance, Interpretability, AI Security*

My research centers on evaluation reliability: whether the tests we run on AI systems measure what we think they measure, and whether results transfer across deployment contexts. I traced a production reproducibility failure where identical code and data produced silently incompatible confidence scores across environments which is the kind of invisible failure that motivates my broader interest in information provenance (model collapse, training-data contamination) and interpretability (verifying model behavior rather than assuming it from outputs). My entry point into this thinking was a prior research project using bioluminescence imaging to locate tumors non-invasively, where I saw directly how much gets lost between what a system detects and what a practitioner can actually trust.

Skills

Programming & Tools: Python, Java, SQL, PostgreSQL, React, REST APIs, GitHub, MATLAB, Wireshark, GNS3

AI Safety & ML: LLM evaluation and benchmarking, RAG pipeline security, prompt injection detection, embedding model analysis, responsible retrieval design

Security: VAPT, network analysis, OWASP, IDOR, CWE documentation

Libraries: ChromaDB, spaCy, BERT, Sentence-Transformers, Scikit-learn, NLTK, Pandas, LangChain, Plotly

Experience

Product Manager (Promoted from SWE after 2 months), Vidzai, Remote, India Nov 2024 - Present

- Built a human-in-the-loop video generation pipeline converting LLM-summarized text into structured JSON and dynamically rendered HTML templates, enabling rapid on-demand video generation for interactive Q&A and high-volume educational content; reducing production cost and hallucination risk versus traditional video generation while enforcing visual consistency at scale.
- Drove 0-to-1 launch of AI-native EdTech platform (Skillr) across a four-sided marketplace; scaled to 4,000+ users and 91%+ session engagement by translating learner pain points into platform features.
- Diagnosed DAU drop to 20 via user interviews across 8 cohorts (200 users); implemented onboarding improvements that drove recovery to 111 next-day and sustained session engagement.
- Led LLM migration evaluation comparing 10 models across latency, multilingual accuracy, and token economics; led developer enablement sessions on AI-assisted coding standards (Cursor, unit testing) across the engineering team.
- Benchmarked 3 embedding models on a multilingual corpus, identified F1-optimal retrieval threshold achieving 100% precision/recall, and optimized RAG pipelines across 3 deployments through failure mode analysis and OOD query rejection to ensure responsible retrieval behavior, cutting response time from 8-10s to 2-4s.

- Secured 2 Fortune 500 enterprise partnerships and advanced 4 opportunities by leading live demos for IIT Guwahati, Intel, Microsoft, and C suite executives via remote demo for the international Bancassurance conference (Morocco).
- Conducted production VAPT; identified 8 critical/high/medium vulnerabilities including IDOR across 3 endpoints, insufficient session expiration (CWE-613), wildcard CORS (CWE-942), and cross-subdomain PII cookie leakage.
- Launched 2 AI-native learning features (LENS: simulation-based exploration; AURA: psychometric-adaptive mentoring) and scaled video curriculum 80% beyond target by writing 106 scripts for Fortune 500 demo readiness.
- Built company-wide knowledge infrastructure (61 Confluence docs) spanning architecture, security, and implementation specs.

Product Operations Associate, *Credent Life Sciences, Remote, India*

Sep 2023 - Oct 2024

- Led platform migration to self-managed infrastructure, achieving 70% cost reduction while implementing MFA, usage logging, and security hardening across the stack.

Software Engineer Intern, *Cisco, Remote, USA*

Jul 2022 - Sep 2022

- Built ML-based anomaly detection system flagging Splunk log discrepancies and routing alerts to engineering teams, replacing a 4-step manual escalation chain.

Projects

ChromaDB Reproducibility Investigation, *Python, ChromaDB, LangChain*

2026

- Traced a 289% score discrepancy across local and shared environments to a silent backend migration.
- Eliminated 8 layers of the retrieval stack (distance calculation, formula selection and implementation, embedding model output) through direct empirical testing.
- Published the full investigation publicly with data and diagrams

RAG Security Scanner, *Python, BERT, spaCy, ChromaDB*

In Progress

- ML-powered tool detecting prompt injection and PII leakage in RAG pipelines using trained classifiers and NER-based token classification. Outputs CWE-classified remediation reports.

Structured LLM Evaluation Framework, *Python*

In Progress

- Constraint-following benchmark with a 3×3 difficulty matrix (constraint count × conflict level), three-layer scoring (automated binary + LLM-as-judge + blind human rating), and a product decision framework mapping reliability against cost. Designed as a living benchmark triggered by model updates.

Education

Master of Computer Science, *University of California, Irvine*

Sep 2021 - Mar 2023

- Contributed to biomedical AI research analyzing 15,000+ patient tumor samples for early cancer detection.
- Relevant coursework: Machine Learning & Data Mining, Introduction to AI, Computer Networking, Computer Security.

Bachelor of Technology, Computer Science & Engineering, *VIT, Bhopal*

Sep 2017 - Jul 2021

- CGPA: 9.28/10 - Top 5 in class of 175. Won Best Manager competition (out of 100 candidates).
- Founded Public Speaking Society (80+ members); Head of Anchoring Committee